

Assignment 4

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For this assignment, I used one mutex and several semaphores to keep the participant and judge threads synchronized. The mutex protects the shared variables so that only one thread can update them at a time, while the semaphores control when each thread is allowed to continue. Participants are only allowed to enter when space is available, the judge waits until enough participants have entered, participants wait until the judge chooses a winner, and the judge waits until everyone has reacted before starting the next lottery. This keeps the shared data organized and prevents race conditions.

My synchronization ensures that the correct order of prints because each step must finish before the next one can begin. The judge cannot start a lottery until all participants have entered, participants cannot react until the winning ticket has been announced, and the next lottery cannot begin until everyone has finished reacting and the judge has thanked the participants. After each round, the shared variables are reset so the next lottery starts with a clean state and follows the same sequence.